Android Practical Practice Code

1. **Create a login screen with EditText fields for username and password, and a Button to submit. Include validation for empty fields.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="15dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity_kt

```
package com.example.loginapp
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast

class MainActivity : AppCompatActivity() {

    lateinit var username: EditText
    lateinit var password: EditText
    lateinit var loginBtn: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        username = findViewById(R.id.etUsername)
        password = findViewById(R.id.etPassword)
        loginBtn = findViewById(R.id.btnLogin)

        loginBtn.setOnClickListener {

            val user = username.text.toString()
            val pass = password.text.toString()

            if (user.isEmpty()) {
                username.error = "Username required"
                username.requestFocus()
            } else if (pass.isEmpty()) {
                password.error = "Password required"
                password.requestFocus()
            } else {
                Toast.makeText(this, "Login Successful!", Toast.LENGTH_SHORT).show()
            }
        }
    }
}
```

2. **Create an android application to display Alert Dialog on pressing the Back button.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Press Back Button to see Alert Dialog"
        android:textSize="18sp"/>

</LinearLayout>
```

MainActivity_kt

```
package com.example.alertback

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import androidx.appcompat.app.AlertDialog

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }

    // This method runs when Back button is pressed
    override fun onBackPressed() {

        val builder = AlertDialog.Builder(this)
        builder.setTitle("Exit App")
        builder.setMessage("Are you sure you want to exit?")
```

```
        builder.setPositiveButton("Yes") { _, _ ->
            finish()   // closes the app
        }

        builder.setNegativeButton("No") { dialog, _ ->
            dialog.dismiss()  // closes dialog only
        }

        builder.setCancelable(false)

        val alertDialog = builder.create()
        alertDialog.show()
    }
}
```

3. **Create an android application to pass the data from one activity to another activity in the same application using intent.**

Activity_xml (1st UI screen)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:text="Send Data"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity_kt (send data logic)

```
package com.example.intentdata
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText

class MainActivity : AppCompatActivity() {

    lateinit var name: EditText
    lateinit var sendBtn: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        name = findViewById(R.id.etName)
        sendBtn = findViewById(R.id.btnSend)

        sendBtn.setOnClickListener {

            val userName = name.text.toString()

            val intent = Intent(this, SecondActivity::class.java)
            intent.putExtra("username", userName)

            startActivity(intent)
        }
    }
}
```

Activity_xml (2nd UI Screen)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"/>

</LinearLayout>
```

Mainactivity_kt (Received data)

```
package com.example.intentdata
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.TextView

class SecondActivity : AppCompatActivity() {

    lateinit var result: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)

        result = findViewById(R.id.tvResult)

        val name = intent.getStringExtra("username")

        result.text = "Welcome $name"
    }
}
```

Manifest_xml

<activity android:name=".SecondActivity" />

4. **Create a user interface with a TextView, Button, and ImageView. When the button is clicked, the image should be changed dynamically**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Dynamic Image Change"
        android:textSize="20sp"
        android:textStyle="bold"/>

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/image1"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnChange"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Image"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity_kt

```
package com.example.dynamicimage
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.ImageView

class MainActivity : AppCompatActivity() {

    lateinit var imageView: ImageView
    lateinit var changeBtn: Button

    var isImageOne = true

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        imageView = findViewById(R.id.imageView)
        changeBtn = findViewById(R.id.btnChange)

        changeBtn.setOnClickListener {

            if (isImageOne) {
                imageView.setImageResource(R.drawable.image2)
                isImageOne = false
            } else {
                imageView.setImageResource(R.drawable.image1)
                isImageOne = true
            }
        }
    }
}
```

5. **Create an android application, with one button and image view also set the background image in the application, the image should be changed on each click of the button (Use two images).**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/bg_image">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/pic1"
        android:layout_centerInParent="true"/>

    <Button
        android:id="@+id/btnChange"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Image"
        android:layout_below="@id/imageView"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

</RelativeLayout>
```

MainActivity_kt

```
package com.example.imagechange
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.ImageView
```

```
class MainActivity : AppCompatActivity() {

    lateinit var imageView: ImageView
    lateinit var button: Button

    var isFirstImage = true

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        imageView = findViewById(R.id.imageView)
        button = findViewById(R.id.btnChange)

        button.setOnClickListener {

            if (isFirstImage) {
                imageView.setImageResource(R.drawable.pic2)
                isFirstImage = false
            } else {
                imageView.setImageResource(R.drawable.pic1)
                isFirstImage = true
            }
        }
    }
}
```

6. **Design a form with a RadioGroup for selecting the gender (Male/Female/Other). Include a Submit button that shows the selected gender in a TextView.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Select Gender"
    android:textSize="20sp"
    android:textStyle="bold"/>

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp">

    <RadioButton
        android:id="@+id/rbMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"/>

    <RadioButton
        android:id="@+id/rbFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"/>

    <RadioButton
        android:id="@+id/rbOther"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Other"/>

</RadioGroup>

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    android:layout_marginTop="20dp"/>
```

```
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity_kt

```
package com.example.radiogroupdemo
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.RadioGroup
import android.widget.TextView
import android.widget.Toast

class MainActivity : AppCompatActivity() {

    lateinit var radioGroup: RadioGroup
    lateinit var submitBtn: Button
    lateinit var result: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        radioGroup = findViewById(R.id.radioGroup)
        submitBtn = findViewById(R.id.btnSubmit)
        result = findViewById(R.id.tvResult)

        submitBtn.setOnClickListener {

            val selectedId = radioGroup.checkedRadioButtonId

            if (selectedId == -1) {
```

```
            Toast.makeText(this, "Please select gender", Toast.LENGTH_SHORT).show()
        } else {
            val selectedGender = findViewById<android.widget.RadioButton>(selectedId)
            result.text = "Selected Gender: ${selectedGender.text}"
        }
    }
  }
}
```

7. **Create an app that displays an ImageView. When the user clicks the ImageView, show a Toast with the message "Image clicked!".**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <ImageView
    android:id="@+id/imageView"
    android:layout_width="200dp"
    android:layout_height="200dp"
    android:src="@drawable/sample_image"
    android:contentDescription="Sample Image"/>

</LinearLayout>
```

MainActivity_kt

```
package com.example.imageclicktoast
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.ImageView
import android.widget.Toast
```

```
class MainActivity : AppCompatActivity() {

    lateinit var imageView: ImageView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        imageView = findViewById(R.id.imageView)

        imageView.setOnClickListener {
            Toast.makeText(this, "Image clicked!", Toast.LENGTH_SHORT).show()
        }
    }
}
```

8. **Design a simple login form using ConstraintLayout that has:**
   - **Username and Password fields aligned vertically.**
   - **Login button centered horizontally below the fields.**
   - **Forgot Password? link aligned to the right of the password field.**
   - **Task: Position all the elements using constraints in the ConstraintLayout.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <!-- Username Field -->
    <EditText
        android:id="@+id/etUsername"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Username"
        app:layout_constraintTop_toTopOf="parent"
```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

    <!-- Password Field -->
    <EditText
        android:id="@+id/etPassword"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginTop="16dp"
        app:layout_constraintTop_toBottomOf="@id/etUsername"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

    <!-- Forgot Password Link -->
    <TextView
        android:id="@+id/tvForgot"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Forgot Password?"
        android:textColor="#1976D2"
        android:layout_marginTop="8dp"
        app:layout_constraintTop_toBottomOf="@id/etPassword"
        app:layout_constraintEnd_toEndOf="@id/etPassword"/>

    <!-- Login Button -->
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="24dp"
        app:layout_constraintTop_toBottomOf="@id/tvForgot"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

9. **Design a screen with two buttons: Start and Stop. When the Start button is clicked, start a Timer (display the time in seconds). When the Stop button is clicked, stop the timer.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0"
        android:textSize="40sp"
        android:textStyle="bold"/>

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnStop"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

Mainactivity_kt

```
package com.example.timerapp
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.os.Handler
import android.widget.Button
import android.widget.TextView

class MainActivity : AppCompatActivity() {

    lateinit var timerText: TextView
    lateinit var startBtn: Button
    lateinit var stopBtn: Button

    var seconds = 0
    var isRunning = false

    val handler = Handler()

    val runnable = object : Runnable {
        override fun run() {
            if (isRunning) {
                seconds++
                timerText.text = seconds.toString()
                handler.postDelayed(this, 1000) // repeat every 1 sec
            }
        }
    }

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        timerText = findViewById(R.id.tvTimer)
        startBtn = findViewById(R.id.btnStart)
        stopBtn = findViewById(R.id.btnStop)

        startBtn.setOnClickListener {
            isRunning = true
```

```
            handler.post(runnable)
        }

        stopBtn.setOnClickListener {
            isRunning = false
        }
    }
}
```

10. **Create a BroadcastReceiver that listens for the Android Intent ACTION_BATTERY_CHANGED broadcast and shows a Toast displaying the current battery percentage.**

Activity_xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Battery Receiver Running"
        android:textSize="20sp"/>

</LinearLayout>
```

Battery_kt

```
package com.example.batteryreceiver
import android.content.BroadcastReceiver
import android.content.Context
import android.content.Intent
import android.os.BatteryManager
import android.widget.Toast
```

```
class BatteryReceiver : BroadcastReceiver() {

    override fun onReceive(context: Context, intent: Intent) {

        val level = intent.getIntExtra(BatteryManager.EXTRA_LEVEL, -1)
        val scale = intent.getIntExtra(BatteryManager.EXTRA_SCALE, -1)

        val batteryPct = (level * 100) / scale

        Toast.makeText(context, "Battery: $batteryPct%", Toast.LENGTH_SHORT).show()
    }
}
```

MainActivity_kt

```
package com.example.batteryreceiver
import android.content.Intent
import android.content.IntentFilter
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle

class MainActivity : AppCompatActivity() {

    lateinit var batteryReceiver: BatteryReceiver

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        batteryReceiver = BatteryReceiver()

        val filter = IntentFilter(Intent.ACTION_BATTERY_CHANGED)
        registerReceiver(batteryReceiver, filter)
    }
    override fun onDestroy() {
        super.onDestroy()
        unregisterReceiver(batteryReceiver)
    }
}
```